

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location IA facility IA_way_1454813846 -- station KDSM, America/Chicago
Committed pair OSM 1454813846 -> 1368776076
OSM way 1454813846 / OSM way 1368776076
Facade gap 175.6 m between the two halls
Weather record 43,810 real hourly records from KDSM
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise 0.3746 C at 180 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2022-11-10 (crossing)
Plant limit 18.0 C
Notice required 3 h
Forecast skill 0.50 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 2 mode changes per day
Minimum dwell 3 h
Max dew point gate off

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 12 of 24 hours with 1 mode change. That is 6 more than the reactive incumbent, at 0 unsafe hours against its 3. 3 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 12 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent 6 of 24 hours free cooling, 2 mode change(s), 3 unsafe hour(s)
3 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	16.341	18.0	16.348	1.157
01	mechanical	yes	switch budget	16.639	18.0	16.909	1.067
02	mechanical	yes	switch budget	16.932	18.0	17.451	1.019
03	mechanical	no	dry-bulb	18.095	18.0	19.221	0.946
04	mechanical	no	dry-bulb	18.301	18.0	19.192	1.110
05	mechanical	no	dry-bulb	18.576	18.0	19.177	1.010
06	mechanical	no	dry-bulb	19.640	18.0	19.177	0.885

07	mechanical	no	dry-bulb	20.284	18.0	19.221	1.274
08	mechanical	no	dry-bulb	21.654	18.0	19.727	1.589
09	mechanical	no	dry-bulb	22.810	18.0	19.755	1.915
10	mechanical	no	dry-bulb	24.193	18.0	20.848	2.042
11	mechanical	no	dry-bulb	18.522	18.0	9.440	1.966
12	FREE	yes	-	17.409	18.0	8.330	1.712
13	FREE	yes	-	16.933	18.0	7.220	1.601
14	FREE	yes	-	10.748	18.0	6.670	1.457
15	FREE	yes	-	9.444	18.0	6.110	1.342
16	FREE	yes	-	7.684	18.0	5.560	1.218
17	FREE	yes	-	6.357	18.0	4.440	1.223
18	FREE	yes	-	4.892	18.0	3.330	1.061
19	FREE	yes	-	4.106	18.0	2.220	1.423
20	FREE	yes	-	2.646	18.0	1.110	1.255
21	FREE	yes	-	1.504	18.0	-0.610	1.468
22	FREE	yes	-	0.450	18.0	-1.720	1.374
23	FREE	yes	-	-0.339	18.0	-2.220	1.223

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.341 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.639 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.932 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.095 C against a 18.0 C limit, so it fails by 0.095 C. It would take a limit 0.095 C higher, or a bound 0.095 C tighter, to change this hour.

04:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.301 C against a 18.0 C limit, so it fails by 0.301 C. It would take a limit 0.301 C higher, or a bound 0.301 C tighter, to change this hour.

05:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.575 C against a 18.0 C limit, so it fails by 0.575 C. It would take a limit 0.575 C higher, or a bound 0.575 C tighter, to change this hour.

06:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.640 C against a 18.0 C limit, so it fails by 1.640 C. It would take a limit 1.640 C higher, or a bound 1.640 C tighter, to change this hour.

07:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.284 C against a 18.0 C limit, so it fails by 2.284 C. It would take a limit 2.284 C higher, or a bound 2.284 C tighter, to change this hour.

08:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.653 C against a 18.0 C limit, so it fails by 3.653 C. It would take a limit 3.653 C higher, or a bound 3.653 C tighter, to change this hour.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.810 C against a 18.0 C limit, so it fails by 4.810 C. It would take a limit 4.810 C higher, or a bound 4.810 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.193 C against a 18.0 C limit, so it fails by 6.193 C. It would take a limit 6.193 C higher, or a bound 6.193 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.522 C against a 18.0 C limit, so it fails by 0.522 C. It would take a limit 0.522 C higher, or a bound 0.522 C tighter, to change this hour.

12:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.409 C, which is 0.591 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.712 C, and the margin is measured, not chosen: 1.518 C of group-conditional forecast error for this hour of day, plus 0.1939 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

13:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.933 C, which is 1.067 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.601 C, and the margin is measured, not chosen: 1.333 C of group-conditional forecast error for this hour of day, plus 0.2675 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

14:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.748 C, which is 7.252 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.457 C, and the margin is measured, not chosen: 1.184 C of group-conditional forecast error for this hour of day, plus 0.2734 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

15:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.444 C, which is 8.556 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.342 C, and the margin is measured, not chosen: 1.063 C of group-conditional forecast error for this hour of day, plus 0.2791 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

16:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.684 C, which is 10.316 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.218 C, and the margin is measured, not chosen: 1.034 C of group-conditional forecast error for this hour of day, plus 0.1843 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

17:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 6.357 C, which is 11.643 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.223 C, and

the margin is measured, not chosen: 0.976 C of group-conditional forecast error for this hour of day, plus 0.2472 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 4.892 C, which is 13.108 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.061 C, and the margin is measured, not chosen: 0.888 C of group-conditional forecast error for this hour of day, plus 0.1722 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 4.106 C, which is 13.894 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.423 C, and the margin is measured, not chosen: 1.206 C of group-conditional forecast error for this hour of day, plus 0.2164 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 2.646 C, which is 15.354 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.255 C, and the margin is measured, not chosen: 1.109 C of group-conditional forecast error for this hour of day, plus 0.1462 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 1.504 C, which is 16.496 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.468 C, and the margin is measured, not chosen: 1.324 C of group-conditional forecast error for this hour of day, plus 0.1440 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 0.450 C, which is 17.550 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.374 C, and the margin is measured, not chosen: 1.174 C of group-conditional forecast error for this hour of day, plus 0.1999 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is -0.339 C, which is 18.339 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.223 C, and the margin is measured, not chosen: 1.006 C of group-conditional forecast error for this hour of day, plus 0.2164 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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